Temporal Foam and Envelopes (TFE): Emergent Space, Measurement, and Superdeterministic Correlations

A Comprehensive Proposal with Simulations

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Abstract

We develop a self-contained framework in which "space" is not fundamental but emerges from a fully connected temporal substrate (*temporal foam*). Structure is imposed by *envelopes*, represented by operators that dephase or confine dynamics, yielding effective locality, finite signalling speed, and one-dimensional preferred paths we call *timelines*. Particles are excitations that propagate at constant intrinsic speed along timelines; all apparent speeding, slowing, and reversal in a spatial readout are geometric projection effects. We show how measurement corresponds to envelope-induced phase locking (einselection), how mass arises from spectral gaps, transverse confinement, or internal mixing, and how a superdeterministic variant arises naturally from clock-phase synchronization in the foam, violating measurement independence while preserving locality and no-signalling. We present simple simulations: motion on line/circle/spiral timelines; a branch node with a unitary splitter and measurement-induced locking; and a CHSH test under the clock-phase model exhibiting $S(\varepsilon) = (1-\varepsilon)2\sqrt{2} + 2\varepsilon$. We conclude with testable predictions and a roadmap for further work.

Contents

1 Introduction

Nonlocal quantum correlations, decoherence, and the success of holographic ideas suggest that spacetime locality may be emergent from deeper informational structure. We propose a compact formalism—*Temporal Foam + Envelopes (TFE)*—that makes this idea concrete and readily simulable. The central ingredients are:

(i) a fully connected "foam" of micro-events with complex couplings;

(ii) *envelopes* (operators) that sculpt dynamics by dephasing or confining;

(iii) emergent coordinates extracted spectrally from the envelope-filtered couplings;

(iv) timelines (1-D subspaces) supporting constant intrinsic-speed propagation;

(v) measurement as strong envelope-induced phase locking (einselection);

(vi) a superdeterministic variant realized as clock-phase synchronization in the foam.

2 The TFE Formalism

2.1 Primitives

Let $\mathcal{V} = \{1, \ldots, N\}$ be micro-events and $\mathcal{H} = \mathrm{span}\{|i, \sigma\rangle : i \in \mathcal{V},\ \sigma \in \Sigma\}$ the Hilbert space (internal labels σ are optional). A complex fully connected kernel $K \in \mathbb{C}^{N\times N}$ is split into its Hermitian and skew parts

$$H = \tfrac{1}{2}(K + K^\dagger), \qquad A = \tfrac{1}{2i}(K - K^\dagger). \tag{1}$$

The intrinsic time parameter is $\tau \in \mathbb{R}$.

2.2 Open-system dynamics with envelopes

States evolve under a GKLS generator

$$\partial_\tau \rho = -\frac{i}{\hbar}[H, \rho] + \sum_\alpha \gamma_\alpha \Big(L_\alpha \rho L_\alpha^\dagger - \tfrac{1}{2}\{L_\alpha^\dagger L_\alpha, \rho\}\Big), \tag{2}$$

where the *envelope operators* L_α encode structure. Two canonical types:

- **Dephasing envelopes** $L_\alpha = E_\alpha = E_\alpha^\dagger \geq 0$ (phase locking, einselection).
- **Confining envelopes** $L_\alpha = \sqrt{V_\alpha}$ (resistance to leave preferred sets).

2.3 Timelines and resistance to leave them

A *timeline* is a smooth family of rank-1 projectors $\{P(s)\}$, with s arclength. Leakage is penalized by

$$L_{\mathcal{T}}(s) = \sqrt{\kappa}\,(I - P(s)), \qquad \kappa > 0. \tag{3}$$

Inside $\mathrm{im}\, P(s)$, propagation is at constant intrinsic speed $ds/d\tau = c$. Apparent nonuniformities in a spatial *readout* $X = \langle \hat{x}, \gamma(s)\rangle$ are purely geometric.

2.4 Branch nodes and unitary splitters

At a node s where a timeline splits into channels $a = 1, \ldots, m$ with projectors P_a, the outgoing amplitudes are related by a unitary $S \in U(m)$:

$$\Psi_{\mathrm{out}} = S\,\Psi_{\mathrm{in}}. \tag{4}$$

Different leakage penalties κ_a implement tunable splitting ratios.

2.5 Emergent locality and speed limit

Define an envelope-filtered effective adjacency K_{eff} (symmetric, nonnegative), and the Laplacian $L_{\mathrm{diff}} = D - K_{\mathrm{eff}}$. The leading eigenvectors of L_{diff} provide an embedding $\Phi : i \mapsto x(i) \in \mathbb{R}^d$ which serves as emergent coordinates. When K_{eff} is banded in x, the unitary part of (**??**) reduces to a local PDE,

$$i\hbar\,\partial_\tau \psi(x) \approx -\frac{\hbar^2}{2\mu}\,\Delta_g \psi(x) + U(x)\psi(x), \tag{5}$$

and Lieb–Robinson-type bounds yield a finite maximal group velocity, identified with c.

3 Photon Sector and Timeline Geometry

Choose a massless band with linear dispersion, $E(k) \approx \hbar c\|k\|$, i.e. $H_{\mathrm{ph}} \approx \hbar c\sqrt{-\Delta_g}$. Confinement to a 1-D curve $x = x_{\mathcal{T}}(s)$ is enforced by a strong transverse envelope.

Readout kinematics. For a readout $X = \mathrm{Re}$, unit-speed motion along the curve yields

$$\dot{X}(\tau) = c\,\frac{dX}{ds} = c\,\langle T(s), \hat{x}\rangle, \qquad T = \partial_s x_{\mathcal{T}}. \tag{6}$$

Thus $X(t)$ speeds up, slows, and reverses solely from the tangent's projection.

3.1 Line, circle, and spiral (simulated)

We simulate three shapes (figs. **??**–**??**):

- **Line:** $x_{\mathcal{T}}(s) = (s, 0)$, $X = s$ and $\dot{X} = c$.
- **Circle:** $x_{\mathcal{T}}(s) = (R\cos\frac{s}{R}, R\sin\frac{s}{R})$, $X = R\cos\frac{s}{R}$ and $\dot{X} = -c\sin\frac{s}{R}$ (reverses at turning points).
- **Archimedean spiral:** $r = a + b\theta$, $X = r\cos\theta$; reversals persist with growing amplitude and net drift each 2π.

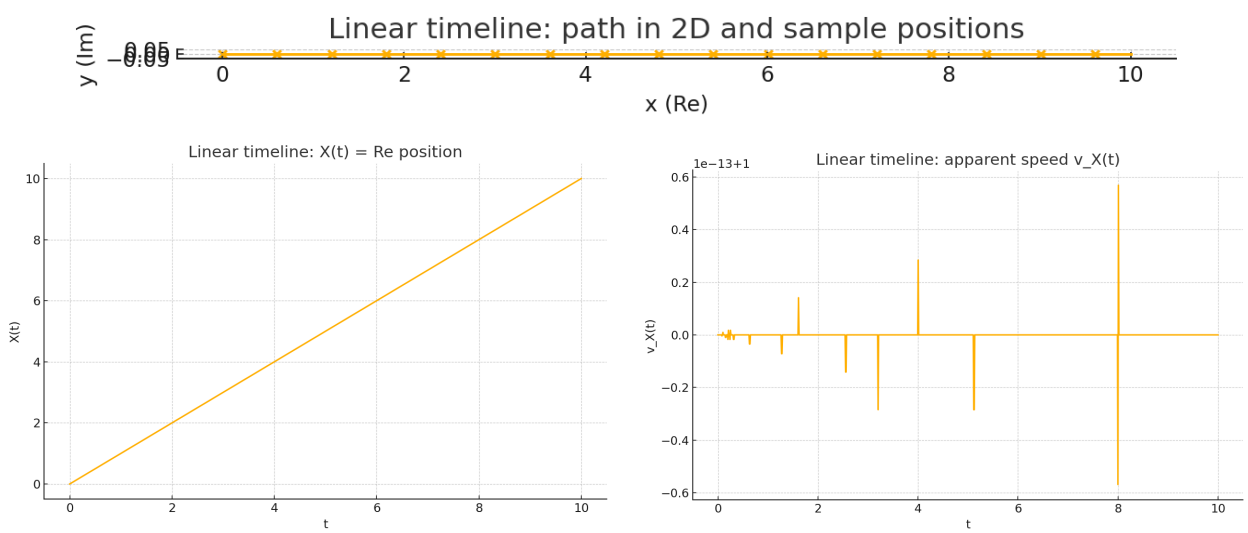

Figure 1: Linear timeline: path (top), readout $X(t)$, and apparent speed $\dot{X}(t)$.

4 Mass in TFE

We present three equivalent definitions.

(i) Spectral gap (primary):

$$mc^2 := E(0) = \hbar\omega(0), \qquad E(k) \approx \sqrt{(mc^2)^2 + (\hbar c\,\|k\|)^2}. \tag{7}$$

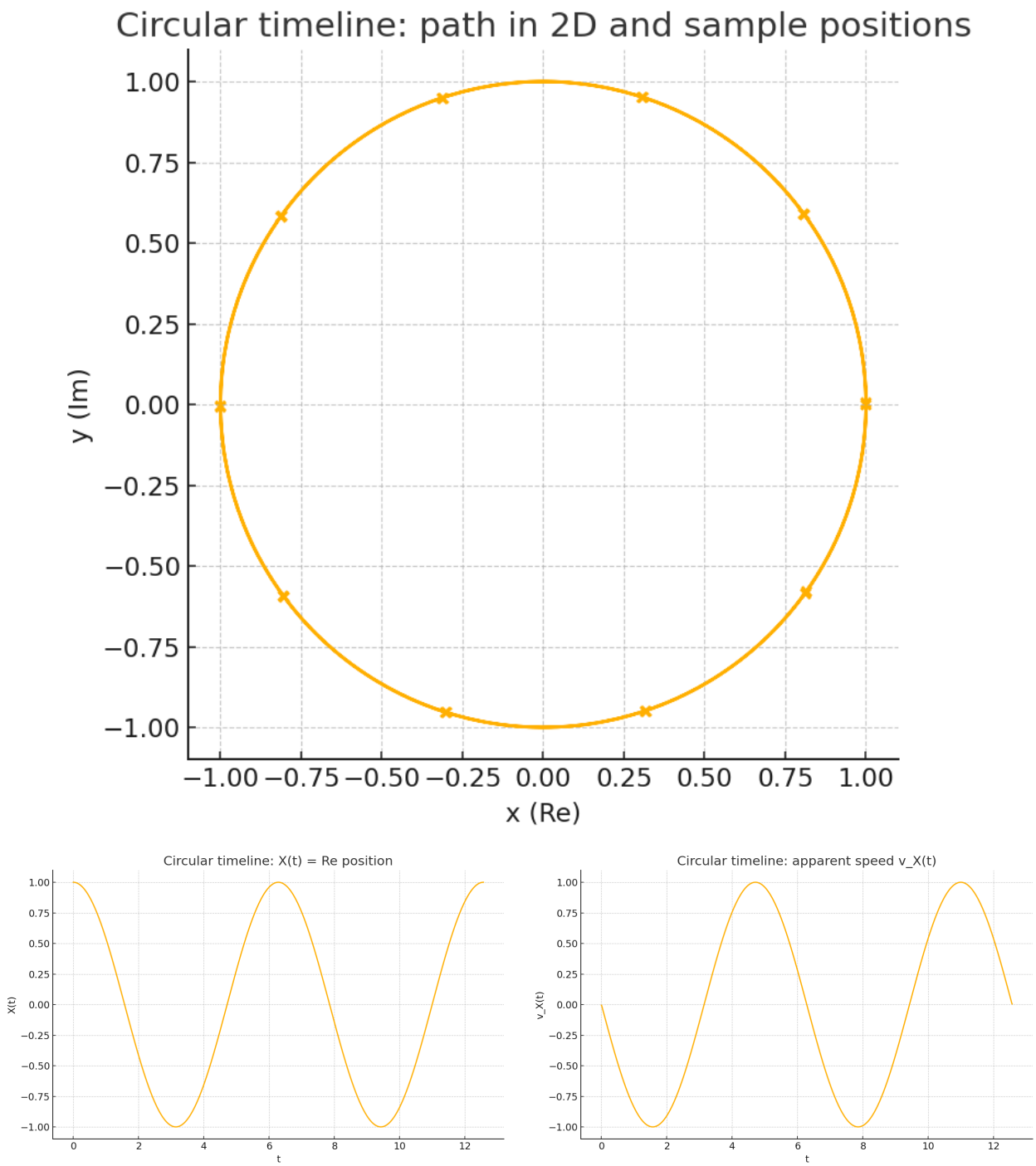

Figure 2: Circular timeline: periodic reversals in the readout arise from projection geometry.

(ii) Confinement/waveguide: For transverse eigenvalues λ_n of the tube,

$$E_n(k) = \sqrt{(\hbar c)^2 \|k\|^2 + (\hbar c)^2 \lambda_n} \quad \Rightarrow \quad m_n c^2 = \hbar c \sqrt{\lambda_n}. \tag{8}$$

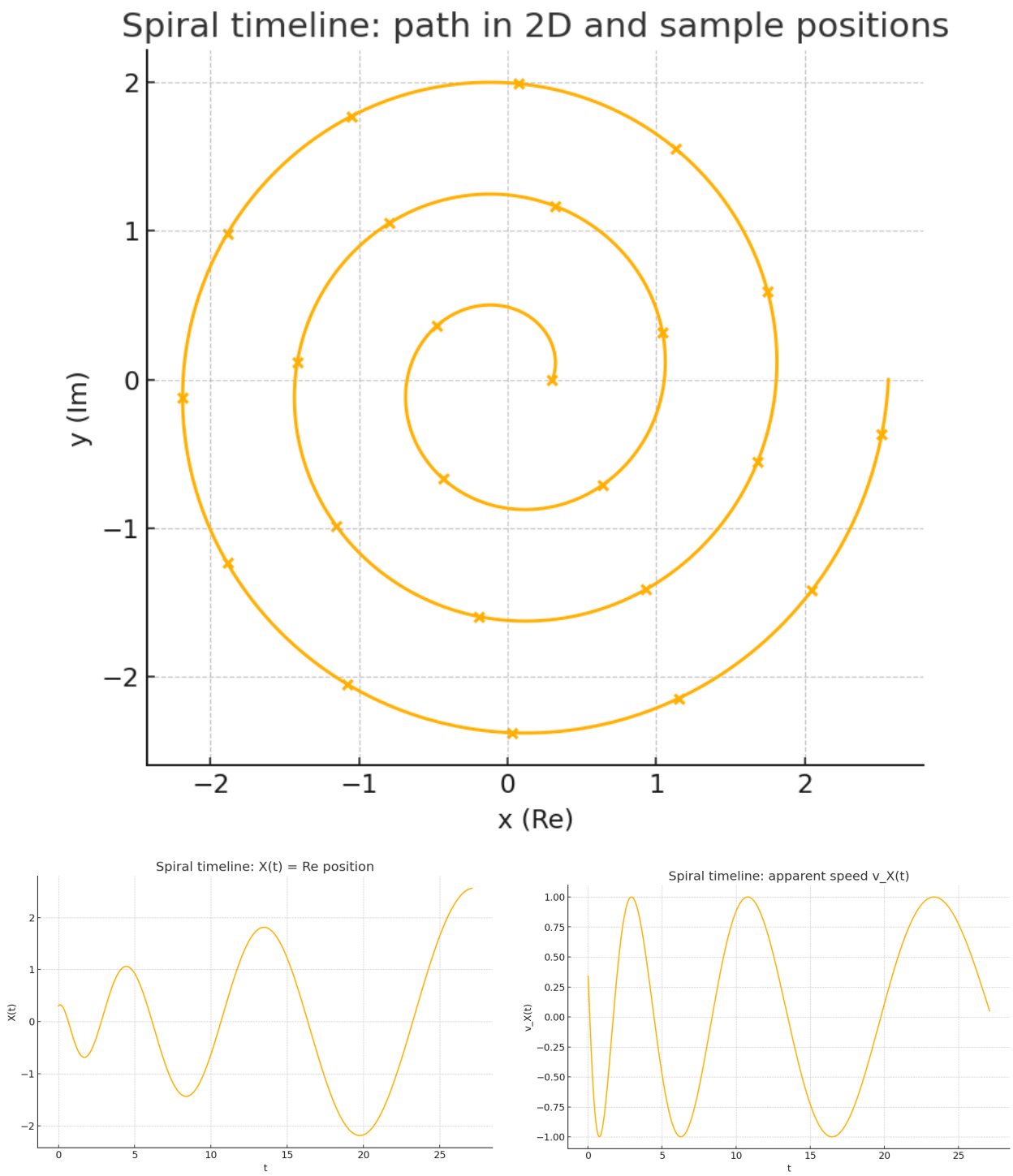

Figure 3: Spiral timeline: reversals with growing amplitude and net drift.

(iii) Internal mixing (Dirac-like): If an envelope mixes internal labels with strength M,

$$H \approx \hbar c\, \boldsymbol{\alpha}\cdot\boldsymbol{k} + \beta\, M, \qquad m = M/c^2. \tag{9}$$

5 Measurement as Envelope-Induced Phase Locking

Strong dephasing envelopes L_{obs} select pointer states and suppress coherences. In a two-time (pre/post) description, probabilities follow the Aharonov–Bergmann–Lebowitz rule while einselection fixes the pointer basis.

6 Branch Nodes and Measurement Locking (Simulation)

We inject a Gaussian wavepacket into a node with a 50/50 unitary splitter S feeding (A) a line and (B) a circle. Without measurement, ensemble $X(t)$ is a weighted sum of the two geometries. Turning on strong which-path dephasing at t_{lock} locks the readout to a single pointer path (fig. **??**).

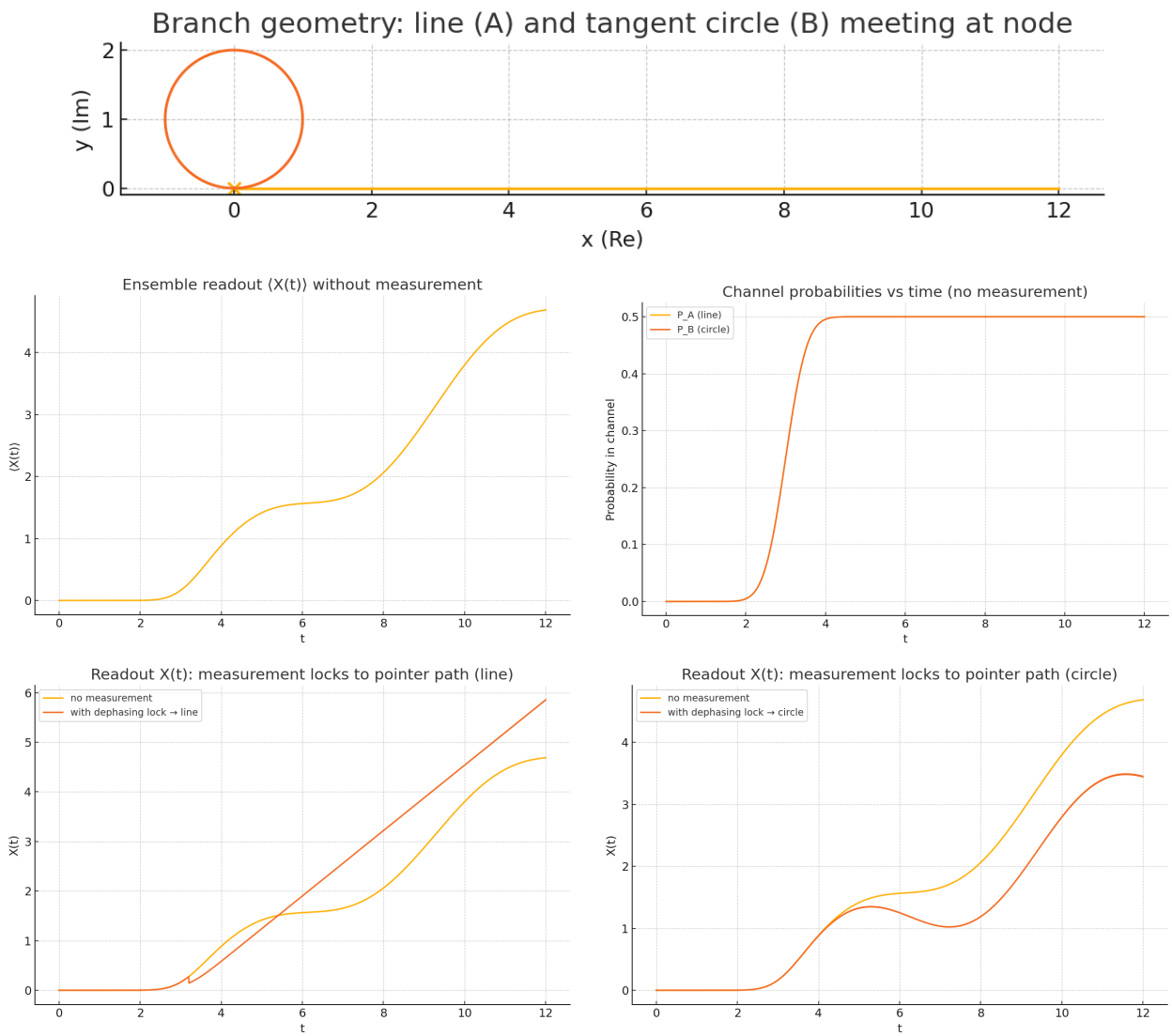

Figure 4: Branch node: geometry; ensemble readout without measurement; channel probabilities; and readouts after dephasing lock to (left) line and (right) circle.

7 Superdeterminism as Temporal Synchronization

7.1 Hidden variable and deterministic outcomes

Let the hidden variable be a global *clock phase* $\lambda \in [0, 2\pi)$. Outcomes for settings a, b are local and deterministic:

$$A(a, \lambda) = \mathrm{sgn}\big(\cos(\lambda - a)\big), \qquad B(b, \lambda) = -\mathrm{sgn}\big(\cos(\lambda - b)\big). \tag{10}$$

7.2 Measurement independence is violated

The setting generators are themselves functions of synchronized envelope structure; therefore $\rho(\lambda \,|\, a, b) \neq \rho(\lambda)$. Choosing a piecewise-uniform $\rho(\lambda \,|\, a, b)$ concentrated on sectors where $AB = \pm 1$ exactly reproduces the quantum correlator $E(a, b) = -\cos|a - b|$.

7.3 Desynchronization knob and CHSH

Let $\varepsilon \in [0, 1]$ be the fraction of trials where λ is drawn from an *independent* uniform distribution (no sync). Then

$$S(\varepsilon) = (1 - \varepsilon)\, 2\sqrt{2} + 2\,\varepsilon. \tag{11}$$

Monte Carlo with $N = 2 \times 10^5$ samples per pair reproduces this trend (fig. **??**).

8 Information Bounds and No-Signalling

Even with global envelope correlations, accessible mutual information across a partition $A|B$ is bottlenecked by the envelope-induced sparsity of K_{eff}:

$$I_{\mathrm{accessible}}(A\!:\!B) \;\leq\; C_{\mathrm{cut}}(A|B) = \sum_{e \in \min\,\mathrm{cut}} C_e, \tag{12}$$

where C_e are link capacities induced by the filtered couplings. This bound underwrites no-signalling while allowing strong correlations.

9 Retrocausality without Paradoxes

Let $\mathcal{E}_+$ and $\mathcal{E}_-$ be forward/backward CP maps induced by envelopes. On the einselected subspace they commute and form a consistent two-time boundary problem:

$$[\mathcal{E}_+, \mathcal{E}_-] = 0, \qquad \mathcal{E}_- \circ \mathcal{E}_+(\rho) = \rho. \tag{13}$$

Retro-influences restrict histories but do not generate contradictions.

10 Predictions and Experimental Knobs

(a) **Synchronization vs dephasing sweep:** Increasing synchronization (lower ε) and lowering dephasing increases S toward $2\sqrt{2}$; the reverse trend lowers S toward 2.

(b) **Indefinite causal order collapse:** As dephasing grows, causal witnesses in quantum-switch experiments should move monotonically into the causally separable region.

(c) **Entanglement–geometry knob:** Varying envelope strength that sparsifies K_{eff} should reduce operational distance and entanglement across a cut.

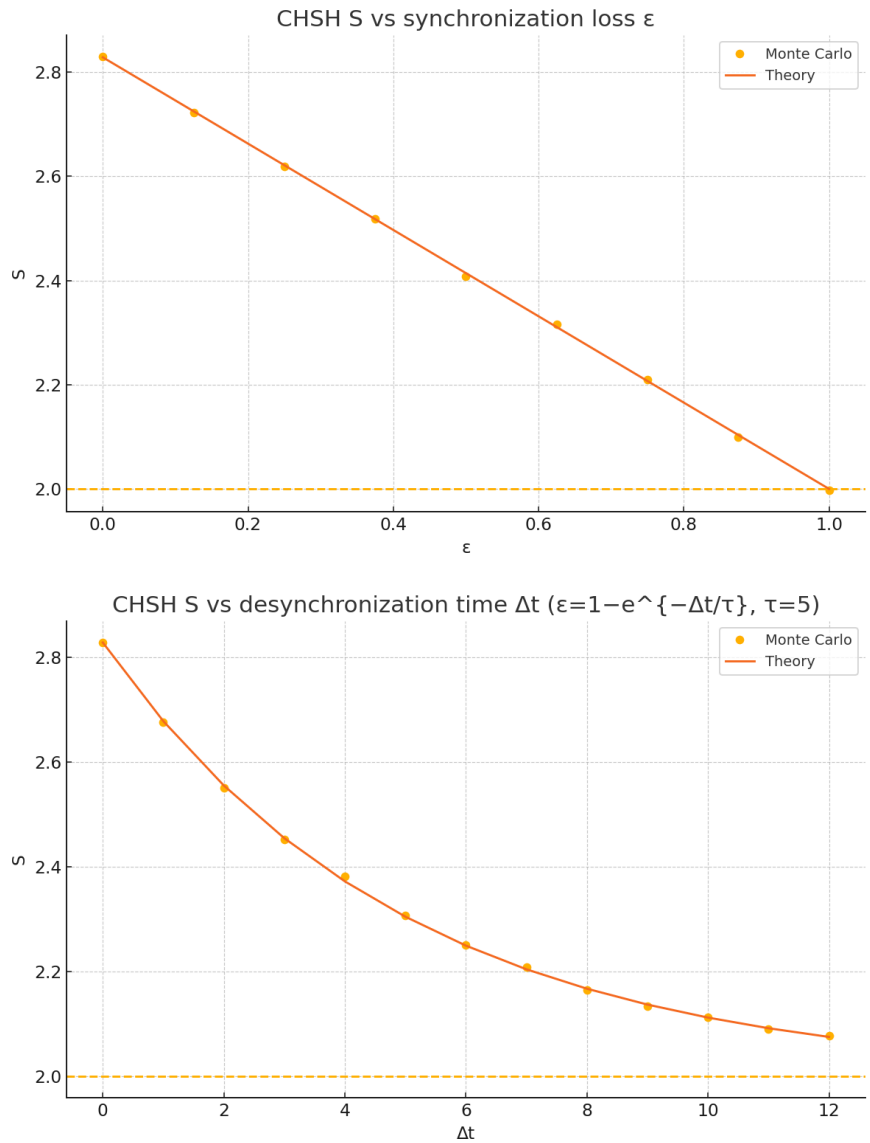

Figure 5: Top: CHSH S vs synchronization loss ε. Bottom: S vs desynchronization time Δt for $\varepsilon(\Delta t) = 1 - e^{-\Delta t/\tau}$ (here $\tau = 5$).

11 Conclusion

TFE provides a small set of primitives that produce emergent space, constant intrinsic-speed propagation along timelines, measurement as phase locking, mass as gap/confinement/mixing, and a superdeterministic variant via temporal synchronization that preserves locality and no-signalling. The simulations illustrate concrete, falsifiable knobs. Future work includes a field-theoretic coarse graining, detailed capacity bounds for K_{eff}, and laboratory tests that co-vary synchronization and dephasing.

Acknowledgments

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References

[1] W. H. Zurek, "Decoherence, einselection, and the quantum origins of the classical," *Rev. Mod. Phys.* **75**, 715 (2003).

[2] J. Maldacena, "The large-N limit of superconformal field theories and supergravity," *Int. J. Theor. Phys.* **38**, 1113 (1999).

[3] S. Ryu and T. Takayanagi, "Holographic derivation of entanglement entropy," *Phys. Rev. Lett.* **96**, 181602 (2006).

[4] J. Maldacena and L. Susskind, "Cool horizons for entangled black holes," *Fortschr. Phys.* **61**, 781 (2013).

[5] Y. Aharonov, P. G. Bergmann, and J. L. Lebowitz, "Time Symmetry in the Quantum Process," *Phys. Rev.* **134**, B1410 (1964).

[6] O. Oreshkov, F. Costa, and Č. Brukner, "Quantum correlations with no causal order," *Nat. Commun.* **3**, 1092 (2012).

[7] E. H. Lieb and D. W. Robinson, "The finite group velocity of quantum spin systems," *Commun. Math. Phys.* **28**, 251 (1972).

[8] F. Pastawski, B. Yoshida, D. Harlow, and J. Preskill, "Holographic quantum error-correcting codes," *JHEP* **06**, 149 (2015).

[9] S. Sachdev and J. Ye, "Gapless spin-fluid ground state in a random, quantum Heisenberg magnet," *Phys. Rev. Lett.* **70**, 3339 (1993).

[10] J. Maldacena and D. Stanford, "Remarks on the Sachdev-Ye-Kitaev model," *Phys. Rev. D* **94**, 106002 (2016).